# Multimodal Stock Trend Predictor: Architecture & Roadmap

*(Note: We are currently freezing all work on Macro and Price data to focus exclusively on solving the Financial News pipeline).*

## Phase 1: News Data Acquisition & Preparation

### Step 1: Sourcing 10 Years of Indian Financial News

Getting a single 10-year CSV of Indian stock news does not exist natively, but we can piece it together perfectly using free, highly-rated databases from Kaggle. Here is where we will get the data:

1. **The Historical Base (2014 - 2020):**
    - **Dataset:** "Indian financial news articles (2003-2020)" on Kaggle.
    - **What it is:** Over 50,000 headlines and descriptions extracted from *Business Standard* and *Moneycontrol*.
2. **The Modern Era (2021 - 2024+):**
    - **Dataset:** "Economic Times Headlines (India) – 2022 to 2025" and "NifSent50: NIFTY 50 Stocks and News Dataset" on Kaggle.
    - **What it is:** These cover recent years with headlines scraped directly from *The Economic Times* and *Investing.com*.

**Action:** You will need to download these CSV files. Once downloaded, we will merge them into one massive spreadsheet containing just `[Date, Headline, Source]`.

### Step 2: Segregating "Stock-Specific" vs "Overall Market" News

As you correctly pointed out, we cannot just throw all news into one bucket. We need to split the news for any given day.

We will use a process called **Keyword/Entity Tagging**:

- **Stock-Specific:** We will maintain a list of keywords for your stock (e.g., if the stock is Reliance, the keywords are "Reliance", "RIL", "Ambani", "Jio"). If an article contains these words, it goes into the **Stock News Bucket**.
- **Overall Market:** If an article mentions "Nifty", "Sensex", "RBI", "Inflation", "Economy", "SEBI", or mentions no specific companies at all, it goes into the **Market News Bucket**.

### Step 3: Preparing the Data for FinBERT

FinBERT expects plain text sentences. To feed the data to FinBERT daily, we will structure it like this:

For every single trading day (e.g., July 15, 2018), we will create TWO text prompts to pass to FinBERT:

1. **Market Prompt:** We take the top 5 "Overall Market" headlines of that day and join them together into a single paragraph.
    - *Example:* "RBI keeps repo rate unchanged. Inflation hits a 3-month high. Sensex drops 400 points amid global selloff."

- *FinBERT Output:* This generates a 768-dimension array representing the **Market Sentiment**.

2. **Stock Prompt:** We take the headlines from the "Stock News Bucket" for that day.
    - *Example:* "Reliance Jio announces new 5G tariff plans. RIL Q3 profits beat street estimates."
    - *FinBERT Output:* This generates a 768-dimension array representing the **Stock Sentiment**.

If there is a day with NO news for your specific stock (which happens often), we simply pass an empty string or a neutral token, and the model will learn to rely entirely on the Overall Market sentiment for that day.

## Conclusion of Phase 1 (News)

By the end of this phase, instead of raw text, every single day in your 10-year timeline will have TWO numerical arrays attached to it:

1. `Market_News_Embedding`
2. `Stock_News_Embedding`

These two arrays will eventually be fed into your model alongside the price data.